

Proofs from the 30062 syllabus

A.Y. 2024 – 2025

NOTE: This document is a collection of screenshots from the official course book*. It has been shared as an additional resource but does not replace the official book. The theorems appear in the same order as in the syllabus.

*Version: Simone Cerreia-Vioglio, Massimo Marinacci, Elena Vigna, Principles of Mathematics for Economics, draft version, September 2024.

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Structures

H8 – Double complement

Proposition 8 $(A^c)^c = A$.

Proof As for Proposition 4, we have to verify an equality between sets. We thus have to prove separately the two inclusions $(A^c)^c \subseteq A$ and $A \subseteq (A^c)^c$. If $a \in (A^c)^c$, then $a \notin A^c$ and therefore $a \in A$. It follows that $(A^c)^c \subseteq A$. Vice versa, if $a \in A$, then $a \notin A^c$ and therefore $a \in (A^c)^c$. We conclude that $A \subseteq (A^c)^c$. ■

*15 – De Morgan's laws

Proposition 15 *Given two subsets A and B of a space S , we have $(A \cup B)^c = A^c \cap B^c$ and $(A \cap B)^c = A^c \cup B^c$.*

Proof We prove only the first law, leaving the second one to the reader. As usual, to prove an equality between sets we have to consider separately the two inclusions that compose it.

(i) $(A \cup B)^c \subseteq A^c \cap B^c$. If $x \in (A \cup B)^c$, then $x \notin A \cup B$, that is, x belongs neither to A nor to B . It follows that x belongs simultaneously to A^c and to B^c and, therefore, to their intersection.

(ii) $A^c \cap B^c \subseteq (A \cup B)^c$. If $x \in A^c \cap B^c$ then $x \notin A$ and $x \notin B$; therefore, x does not belong to their union. ■

H36 – Uniqueness of maximum/minimum

Proposition 36 *A set $A \subseteq \mathbb{R}$ has at most one maximum and one minimum.*

Proof Let $\hat{x}_1, \hat{x}_2 \in A$ be two maxima of A . We show that $\hat{x}_1 = \hat{x}_2$. Since $\hat{x}_1$ is a maximum, we have $\hat{x}_1 \geq x$ for every $x \in A$. In particular, since $\hat{x}_2 \in A$, we have $\hat{x}_1 \geq \hat{x}_2$. Analogously, $\hat{x}_2 \geq \hat{x}_1$ because also $\hat{x}_2$ is a maximum. Therefore, $\hat{x}_1 = \hat{x}_2$. In a similar way, we can prove the uniqueness of the minimum. ■

H111 – Triangle inequality

Corollary 111 *Let $x, y \in \mathbb{R}^n$. Then:*

$$\|x + y\| \leq \|x\| + \|y\| \quad (4.10)$$

Proof We proceed as in the (algebraic) proof of the triangle inequality for the scalar case. Compared to it, we only need to resort to the Cauchy-Schwarz inequality for the very last step that, instead, in the scalar case was trivially true. Note that

$$\begin{aligned} \|x + y\|^2 &\leq (\|x\| + \|y\|)^2 \iff \|x + y\|^2 \leq \|x\|^2 + \|y\|^2 + 2\|x\|\|y\| \\ &\iff x \cdot x + 2(x \cdot y) + y \cdot y \leq x \cdot x + y \cdot y + 2\|x\|\|y\| \\ &\iff x \cdot y \leq \|x\|\|y\| \end{aligned}$$

Since the last inequality follows from the Cauchy-Schwarz inequality, the statement follows too. ■

H128 – Characterization of supremum

Proposition 128 *Let $A \subseteq \mathbb{R}$ be bounded above. We have $a = \sup A$ if and only if*

- (i) $a \geq x$ for every $x \in A$,
- (ii) for every $\varepsilon > 0$, there exists $x \in A$ such that $x > a - \varepsilon$.

Thus, point $a \in \mathbb{R}$ is the supremum of $A \subseteq \mathbb{R}$ if and only if (i) it is an upper bound of A and (ii) in each left neighborhood of a there are elements of A . A similar characterization holds for infima of lower bounded sets by replacing left neighborhoods with right ones.

Proof “Only if”. If $a = \sup A$, (i) is obviously satisfied. Let $\varepsilon > 0$. Since $\sup A > a - \varepsilon$, the point $a - \varepsilon$ is not an upper bound of A . Therefore, there exists $x \in A$ such that $x > a - \varepsilon$.

“If”. Suppose that $a \in \mathbb{R}$ satisfies (i) and (ii). By (i), a is an upper bound of A . By (ii), it is also the least upper bound. Indeed, each $b < a$ can be written as $b = a - \varepsilon$, by setting $\varepsilon = a - b > 0$. Given $b < a$, by (ii) there exists $x \in A$ such that $x > a - \varepsilon = b$. Therefore, b is not an upper bound of A , which implies that there is no upper bound smaller than a . ■

*148– Neighborhoods are open sets

Lemma 148 *Neighborhoods are open sets.*

Proof Let $B_\varepsilon(x_0)$ be a neighborhood of a point $x_0 \in \mathbb{R}^n$. To show that $B_\varepsilon(x_0)$ is open, we have to show that all its points are interior. Let $x \in B_\varepsilon(x_0)$. To prove that x is interior to $B_\varepsilon(x_0)$, let

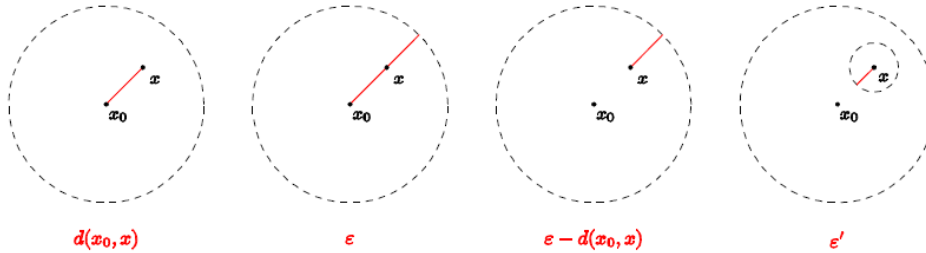
$$0 < \varepsilon' < \varepsilon - d(x, x_0) \quad (5.2)$$

Then $B_{\varepsilon'}(x) \subseteq B_\varepsilon(x_0)$. Indeed, let $y \in B_{\varepsilon'}(x)$. Then

$$d(y, x_0) \leq d(y, x) + d(x, x_0) < \varepsilon' + d(x, x_0) < \varepsilon$$

where the last inequality follows from (5.2). Therefore $B_{\varepsilon'}(x) \subseteq B_\varepsilon(x_0)$, which completes the proof. ■

This proof can be illustrated by the following picture:



H154 – Open/closed sets and their complements

Theorem 154 *A set A in $\mathbb{R}^n$ is open if and only if its complement is closed.*

Proof “Only if”. Let A be open. We show that A^c is closed. Let x be a boundary point of A^c , that is, $x \in \partial A^c$. By definition, x is not an interior point of either A or A^c . Hence, $x \notin \text{int } A$. But, $A = \text{int } A$ because A is open. Therefore $x \notin A$, that is, $x \in A^c$. It follows that $\partial A^c \subseteq A^c$ since $x \in \partial A^c$. Therefore, $\overline{A^c} = A^c$, which proves that A^c is closed.

“If”. Let A^c be closed. We show that A is open. Let x be a point of A . Since $x \notin A^c = \overline{A^c}$, the point x is not a boundary point of A^c . It is, therefore, an interior point of either A or A^c . But, since $x \notin A^c$ implies $x \notin \text{int } A^c$, we have $x \in \text{int } A$. Since x was arbitrarily chosen, we conclude that A is open. ■

H160 – Closed sets and limit points

Corollary 160 *A set A in $\mathbb{R}^n$ is closed if and only if it contains all its limit points.*

Proof Let A be closed. By definition, $\bar{A} = A$ and hence, by Proposition 159, $A \cup A' = A$, that is, $A' \subseteq A$. Vice versa, if $A' \subseteq A$, then obviously $A \cup A' = A$. By Proposition 159, $\bar{A} = A \cup A' = A$. ■

Functions

*221 – Invertibility and monotonicity

Proposition 221 *An increasing function $f : A \subseteq \mathbb{R} \rightarrow \mathbb{R}$ is strictly increasing if and only if*

$$f(x) = f(y) \implies x = y \quad \forall x, y \in A \quad (6.19)$$

that is, if and only if it is injective.

A similar result holds for strictly decreasing functions since f is, clearly, strictly decreasing if and only if $-f$ is strictly increasing. Strictly monotone functions are therefore injective, and so invertible.¹⁸

Proof “Only if”. Let f be strictly increasing and let $f(x) = f(y)$. Suppose, by contradiction, that $x \neq y$, say $x > y$. By (6.17), we have $f(x) \neq f(y)$, which contradicts $f(x) = f(y)$. It follows that $x = y$, as desired.

“If”. Suppose that (6.19) holds. Let f be increasing. We prove that it is also strictly increasing. Let $x > y$. By increasing monotonicity, we have $f(x) \geq f(y)$, but we cannot have $f(x) = f(y)$ because (6.19) would imply $x = y$. Thus $f(x) > f(y)$, as claimed. ■

*864 – Concavity and upper contour sets

Proposition 864 *If $f : C \rightarrow \mathbb{R}$ is concave, all its upper contour sets $(f \geq k)$ are convex.*

Proof Given $k \in \mathbb{R}$, let $(f \geq k)$ be non-empty (otherwise, the result is obvious because empty sets are trivially convex). Let $x, y \in (f \geq k)$ and $\alpha \in [0, 1]$. By the concavity of f ,

$$f(\alpha x + (1 - \alpha)y) \geq \alpha f(x) + (1 - \alpha)f(y) \geq \alpha k + (1 - \alpha)k = k$$

Therefore, $\alpha x + (1 - \alpha)y \in (f \geq k)$. ■

H - Concave functions are quasi-concave

Concave functions are quasi-concave because

$$f(\alpha x + (1 - \alpha)y) \geq \alpha f(x) + (1 - \alpha)f(y) \geq \min\{f(x), f(y)\}$$

while convex functions are quasi-convex. In particular, affine functions are quasi-affine. The converses of these implications are false, as the following example shows.

*1021 – Preservation of optimizers under strictly increasing composition

Proposition 1021 *Let $g : B \subseteq \mathbb{R} \rightarrow \mathbb{R}$ be a strictly increasing function with $\text{Im } f \subseteq B$. The two optimization problems*

$$\max_x f(x) \quad \text{sub } x \in C \quad (22.4)$$

and

$$\max_x (g \circ f)(x) \quad \text{sub } x \in C \quad (22.5)$$

are equivalent, that is, they have the same solutions.

Proof Since g is strictly increasing, by Proposition 224 we have

$$s \geq t \iff g(s) \geq g(t) \quad \forall s, t \in \text{Im } f$$

Thus,

$$\underbrace{f(x)}_s \geq \underbrace{f(y)}_t \iff g(f(x)) \geq g(f(y)) \quad \forall x, y \in A$$

Therefore,

$$\forall x \in C, f(\hat{x}) \geq f(x) \iff \forall x \in C, (g \circ f)(\hat{x}) \geq (g \circ f)(x)$$

This proves that a vector $\hat{x} \in C$ solves problem (22.4) if and only if it solves problem (22.5). ■

*1083 – Fenchel's theorem

Theorem 1083 (Fenchel) *Let $f : C \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ be a concave function defined on a convex set. If the point $\hat{x} \in C$ is a local maximizer, then it is a global maximizer.*

Proof Let $\hat{x} \in C$ be a local maximizer. Take any point $x \in C$ and, for each $t \in (0, 1)$, consider the convex combination $x_t = (1 - t)\hat{x} + tx$. It holds $x_t \rightarrow \hat{x}$ as $t \rightarrow 0^+$. Indeed,

$$\lim_{t \rightarrow 0^{+-}} \|x_t - \hat{x}\| = \lim_{t \rightarrow 0^{+-}} \|(1 - t)\hat{x} + tx - \hat{x}\| = \|x - \hat{x}\| \lim_{t \rightarrow 0^{+-}} t = 0$$

Thus, there is a scalar $\bar{t} > 0$ sufficiently small such that $x_{\bar{t}} \in B_\varepsilon(\hat{x}) \cap C$. As a result, by the concavity of f we have

$$f(\hat{x}) \geq f(x_{\bar{t}}) \geq (1 - \bar{t})f(\hat{x}) + \bar{t}f(x)$$

Thus, $\bar{t}f(\hat{x}) \geq \bar{t}f(x)$ and so $f(\hat{x}) \geq f(x)$. We conclude that $\hat{x}$ is a global maximizer. ■

Sequences & series

*316 – Reciprocals and convergence

Proposition 316 *A sequence $\{x_n\}$, with eventually $x_n > 0$, diverges positively if and only if the sequence $\{1/x_n\}$ converges to zero.*

A dual result holds for negative divergence.¹⁵

Proof “If”. Let $1/x_n \rightarrow 0$. Let $K > 0$. Setting $\varepsilon = 1/K > 0$, by Definition 305 there exists $n_{1/K} \geq 1$ such that $0 < 1/x_n < 1/K$ for every $n \geq n_{1/K}$. Therefore, $x_n > K$ for every $n \geq n_{1/K}$, and by Definition 313 we have $x_n \rightarrow +\infty$.

“Only if”. Let $x_n \rightarrow +\infty$ and let $\varepsilon > 0$. Setting $K = 1/\varepsilon > 0$, by Definition 313 there exists $n_{1/\varepsilon}$ such that $x_n > 1/\varepsilon$ for every $n \geq n_{1/\varepsilon}$. Therefore, $0 < 1/x_n < \varepsilon$ for every $n \geq n_{1/\varepsilon}$ and so $1/x_n \rightarrow 0$. ■

*321 – Uniqueness of the limit

Theorem 321 (Uniqueness of the limit) *A sequence $\{x_n\}$ converges to at most one limit $L \in \overline{\mathbb{R}}$.*

Proof Suppose, by contradiction, that there exist two distinct limits L' and L'' that belong to the set $\overline{\mathbb{R}}$. Without loss of generality, we assume that $L'' > L'$. We consider different cases and show that in each of them we reach a contradiction. So, $L' = L''$ and we conclude that the limit is unique.

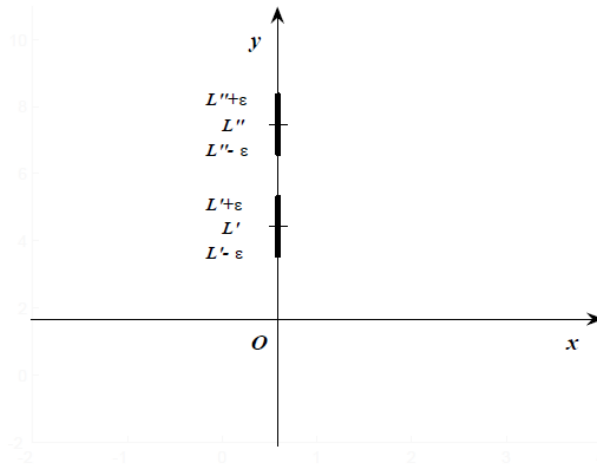
We begin with the case when both L' and L'' are finite, i.e., $L', L'' \in \mathbb{R}$. Take $\varepsilon > 0$ so that

$$\varepsilon < \frac{L'' - L'}{2}$$

Then

$$B_\varepsilon(L') \cap B_\varepsilon(L'') = \emptyset$$

as the reader can verify and the next figure illustrates:



By Definition 306, there exists $n'_\varepsilon \geq 1$ such that $x_n \in B_\varepsilon(L')$ for every $n \geq n'_\varepsilon$, and there exists $n''_\varepsilon \geq 1$ such that $x_n \in B_\varepsilon(L'')$ for every $n \geq n''_\varepsilon$. Setting $n_\varepsilon = \max\{n'_\varepsilon, n''_\varepsilon\}$, we have therefore both $x_n \in B_\varepsilon(L')$ and $x_n \in B_\varepsilon(L'')$ for every $n \geq n_\varepsilon$, i.e., $x_n \in B_\varepsilon(L') \cap B_\varepsilon(L'')$

for every $n \geq n_\varepsilon$. But this contradicts $B_\varepsilon(L') \cap B_\varepsilon(L'') = \emptyset$. We conclude that $L' = L''$, so the limit is unique.

Turn now to the case in which L' is finite and $L'' = +\infty$. For every $\varepsilon > 0$ and every $K > 0$, there exist n_ε and n_K such that

$$L' - \varepsilon < x_n < L' + \varepsilon \quad \forall n \geq n_\varepsilon \quad \text{and} \quad x_n > K \quad \forall n \geq n_K$$

For $n \geq \max\{n_\varepsilon, n_K\}$, we therefore have simultaneously

$$L' - \varepsilon < x_n < L' + \varepsilon \quad \text{and} \quad x_n > K$$

It is now sufficient to take $K = L' + \varepsilon$ to realize that, for $n \geq \max\{n_\varepsilon, n_K\}$, the two inequalities cannot coexist. Also, in this case we reached a contradiction.

The remaining cases can be treated in a similar way and are thus left to the reader. ■

H325 – Convergence and boundedness

Proposition 325 *Each convergent sequence is bounded.*

Proof Suppose $x_n \rightarrow L$. Setting $\varepsilon = 1$, there exists $n_1 > 1$ such that $x_n \in B_1(L)$ for every $n \geq n_1$. Let $M > 0$ be a constant such that

$$M > \max[1, d(x_1, L), \dots, d(x_{n_1-1}, L)]$$

We have $d(x_n, L) < M$ for every $n \geq 1$, i.e., $|x_n - L| < M$ for every $n \geq 1$. This implies that, for all $n \geq 1$,

$$L - M < x_n < L + M$$

Therefore, the sequence is bounded. ■

*326 – Monotonicity and regularity

Theorem 326 *Each monotone sequence is regular. In particular,*

- (i) *it converges if it is bounded;*
- (ii) *it diverges positively if it is increasing and unbounded;*
- (iii) *it diverges negatively if it is decreasing and unbounded.*

Proof Let $\{x_n\}$ be an increasing sequence (the proof for decreasing sequences is similar). It can be either bounded or unbounded above (for sure, it is bounded below because $x_1 \leq x_n$ for every $n \geq 1$). Suppose that $\{x_n\}$ is bounded. We want to prove that it is convergent. Let E be the image of the sequence. By hypothesis, it is a bounded subset of $\mathbb{R}$. By the Least Upper Bound Principle, $\sup E$ exists. Set $L = \sup E$. Let us prove that $x_n \rightarrow L$. Let $\varepsilon > 0$.

Since L is the supremum of E , by Proposition 128 we have: (i) $L \geq x_n$ for every $n \geq 1$, (ii) there exists an element x_{n_ε} of E such that $x_{n_\varepsilon} > L - \varepsilon$. Since $\{x_n\}$ is an increasing sequence, it then follows that

$$L \geq x_n \geq x_{n_\varepsilon} > L - \varepsilon \quad \forall n \geq n_\varepsilon$$

Hence, $x_n \in B_\varepsilon(L)$ for every $n \geq n_\varepsilon$, as desired.

Suppose that $\{x_n\}$ is unbounded above. Then, for every $K > 0$ there exists an element x_{n_K} such that $x_{n_K} > K$. Since $\{x_n\}$ is increasing, we then have $x_n \geq x_{n_K} > K$ for every $n \geq n_K$, so it diverges to $+\infty$. ■

*336 – Limit of the sum

Proposition 336 *Let $x_n \rightarrow L \in \overline{\mathbb{R}}$ and $y_n \rightarrow H \in \overline{\mathbb{R}}$. Then:*¹⁹

(i) $x_n + y_n \rightarrow L + H$, provided that $L + H$ is not an indeterminate form (1.24), of the type

$$+\infty - \infty \quad \text{or} \quad -\infty + \infty$$

Proof (i) Let $x_n \rightarrow L$ and $y_n \rightarrow H$, with $L, H \in \mathbb{R}$. This means that, for every $\varepsilon > 0$, there exist n_1 and n_2 such that

$$L - \varepsilon < x_n < L + \varepsilon \quad \forall n \geq n_1 \quad \text{and} \quad H - \varepsilon < y_n < H + \varepsilon \quad \forall n \geq n_2$$

By adding the inequalities member by member, for every $n \geq n_3 = \max\{n_1, n_2\}$ we have

$$L + H - 2\varepsilon < x_n + y_n < L + H + 2\varepsilon$$

Since 2ε is arbitrary, it follows that $x_n + y_n \rightarrow L + H$.

Now let $x_n \rightarrow L \in \mathbb{R}$ and $y_n \rightarrow +\infty$. This means that, for every $\varepsilon > 0$ and for every $K > 0$, there exist n_1 and n_2 such that

$$L - \varepsilon < x_n < L + \varepsilon \quad \forall n \geq n_1 \quad \text{and} \quad y_n > K \quad \forall n \geq n_2$$

By adding, we have, for every $n \geq n_3 = \max\{n_1, n_2\}$,

$$x_n + y_n > K + L - \varepsilon$$

Since $K + L - \varepsilon > 0$ is arbitrary, it follows that $x_n + y_n \rightarrow +\infty$. The other cases with infinite limit are treated similarly.

*341 – Comparison criterion

Theorem 341 (Comparison criterion) *Let $\{x_n\}$, $\{y_n\}$, and $\{z_n\}$ be three sequences. If, eventually,*

$$y_n \leq x_n \leq z_n \tag{8.40}$$

and

$$\lim y_n = \lim z_n = L \in \overline{\mathbb{R}} \tag{8.41}$$

then

$$\lim x_n = L$$

We can think of $\{x_n\}$ as a convict who is escorted by the two policemen $\{y_n\}$ and $\{z_n\}$ (one on each “side”), so he is forced to go wherever they go.

Proof Suppose $L \in \mathbb{R}$ (we leave to the reader the case $L = \pm\infty$). Let $\varepsilon > 0$. From (8.41) it follows, by Definition 306, that there exists n_1 such that $y_n \in B_\varepsilon(L)$ for every $n \geq n_1$, and there exists n_2 such that $z_n \in B_\varepsilon(L)$ for every $n \geq n_2$. Finally, let n_3 be the position starting from which one has $y_n \leq x_n \leq z_n$. Setting $\bar{n} = \max\{n_1, n_2, n_3\}$, we then have $y_n \in B_\varepsilon(L)$, $z_n \in B_\varepsilon(L)$, and $y_n \leq x_n \leq z_n$ for every $n \geq \bar{n}$. So,

$$L - \varepsilon < y_n \leq x_n \leq z_n < L + \varepsilon \quad \forall n \geq \bar{n}$$

that is, $x_n \in B_\varepsilon(L)$ for every $n \geq \bar{n}$. Hence, $x_n \rightarrow L$ as claimed. ■

H362 – Asymptotic equivalence and the little-o

Proposition 362 *We have*

$$x_n \sim y_n \iff x_n = y_n + o(y_n)$$

In words, two sequences are asymptotic when they are equal, up to a component that is asymptotically negligible with respect to them. This result further clarifies how the relation $\sim$ can be seen as an asymptotic equality. Note that, in view of (8.56) we can equivalently write

$$x_n \sim y_n \iff x_n = y_n [1 + o(1)]$$

Proof “If”. From $x_n = y_n + o(y_n)$ it follows that

$$\frac{x_n}{y_n} = \frac{y_n + o(y_n)}{y_n} = 1 + \frac{o(y_n)}{y_n} \rightarrow 1$$

“Only if”. Let $x_n \sim y_n$. Denoting $z_n = x_n - y_n$, one has that

$$\frac{z_n}{y_n} = \frac{x_n - y_n}{y_n} = \frac{x_n}{y_n} - 1 \rightarrow 0$$

and therefore $z_n = o(y_n)$. ■

H380 – Mengoli series

Example 380 (Mengoli series) The *Mengoli series* is given by:

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \cdots + \frac{1}{n(n+1)} + \cdots = \sum_{n=1}^{\infty} \frac{1}{n(n+1)}$$

Since

$$\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$$

one has that

$$\begin{aligned} s_n &= \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \cdots + \frac{1}{n(n+1)} \\ &= 1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \frac{1}{4} + \cdots + \frac{1}{n} - \frac{1}{n+1} = 1 - \frac{1}{n+1} \rightarrow 1 \end{aligned}$$

Therefore,

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = 1$$

So, the Mengoli series converges and has sum 1. ▲

*382 – Geometric series

Example 382 (Geometric series) The *geometric series* with ratio q is defined by:

$$1 + q + q^2 + q^3 + \cdots + q^n + \cdots = \sum_{n=0}^{\infty} q^n$$

Its character depends on the value of q . In particular, we have that:

$$\sum_{n=0}^{\infty} q^n = \begin{cases} +\infty & \text{if } q \geq 1 \\ \frac{1}{1-q} & \text{if } |q| < 1 \\ \text{irregular} & \text{if } q \leq -1 \end{cases}$$

To verify this, we start by observing that when $q = 1$ we have

$$s_n = \underbrace{1 + 1 + \cdots + 1}_{n+1 \text{ times}} = n + 1 \rightarrow +\infty$$

Let now $q \neq 1$. Since

$$\begin{aligned} s_n - qs_n &= 1 + q + q^2 + q^3 + \cdots + q^n - q(1 + q + q^2 + q^3 + \cdots + q^n) \\ &= 1 + q + q^2 + q^3 + \cdots + q^n - (q + q^2 + q^3 + \cdots + q^{n+1}) = 1 - q^{n+1} \end{aligned}$$

we have

$$(1 - q)s_n = 1 - q^{n+1}$$

and therefore, since $q \neq 1$,

$$s_n = \frac{1 - q^{n+1}}{1 - q}$$

It follows that

$$\sum_{n=0}^{\infty} q^n = \lim_{n \rightarrow \infty} \frac{1 - q^{n+1}}{1 - q}$$

The study of this limit is divided into several cases:

(i) if $-1 < q < 1$, we have $q^{n+1} \rightarrow 0$ and so

$$s_n \rightarrow \frac{1}{1 - q}$$

(ii) if $q > 1$, we have $q^{n+1} \rightarrow +\infty$ and so $s_n \rightarrow +\infty$;

- (iii) if $q = -1$, the partial sums of odd order are equal to zero, while those of even order are equal to 1. The sequence formed by them is hence irregular;
- (iv) if $q < -1$, the sequence $\{q^{n+1}\}$ is irregular and, therefore, so is $\{s_n\}$ as well.

To conclude the study of the geometric series, observe that we can use the recursive definition of partial sums (9.3) to guess and verify (by induction) what are the partial sums of the geometric series. The, highly inspired, guess is that

$$s_n = \frac{1 - q^{n+1}}{1 - q}$$

We verify the guess by induction. For $n = 0$ it is trivially true. Assume it is true for n (induction hypothesis). Then

$$s_{n+1} = s_n + q^{n+1} = \frac{1 - q^{n+1}}{1 - q} + q^{n+1} = \frac{1 - q^{n+1} + q^{n+1} - q^{n+2}}{1 - q} = \frac{1 - q^{(n+1)+1}}{1 - q}$$

as desired.⁵

▲

H383 – Necessary theorem for convergence

Theorem 383 *If the series $\sum_{n=1}^{\infty} x_n$ converges, then $x_n \rightarrow 0$.*

Proof Clearly, we have $x_n = s_n - s_{n-1}$ and, given that the series converges, $s_n \rightarrow S$ as well as $s_{n-1} \rightarrow S$. Therefore, $x_n = s_n - s_{n-1} \rightarrow S - S = 0$. ■

H385 – Regularity theorem

Proposition 385 *Each series with positive terms is either convergent or positively divergent. In particular, it is convergent if and only if it is bounded above.⁷*

Limits of functions & continuity

H539 – Uniqueness of the limit

Theorem 539 (Uniqueness of the limit) *Let $f : A \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ and $x_0 \in A'$. There exists at most a unique $L \in \overline{\mathbb{R}}$ such that $\lim_{x \rightarrow x_0} f(x) = L$.*

Proof Let us suppose, by contradiction, that there exist two different limits $L' \neq L''$. Let $\{x_n\}$ be a sequence in A , with eventually $x_n \neq x_0$, such that $x_n \rightarrow x_0$. By Proposition 536, $f(x_n) \rightarrow L'$ and $f(x_n) \rightarrow L''$, which contradicts the uniqueness of the limit for sequences. It follows that $L' = L''$. ■

H541 – Comparison criterion

Theorem 541 (Comparison criterion) *Let $f, g, h : A \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ and $x_0 \in A'$. If*

$$g(x) \leq f(x) \leq h(x) \quad \forall x \in A \quad (12.31)$$

and

$$\lim_{x \rightarrow x_0} g(x) = \lim_{x \rightarrow x_0} h(x) = L \in \overline{\mathbb{R}} \quad (12.32)$$

then

$$\lim_{x \rightarrow x_0} f(x) = L$$

Again we leave to the reader the easy “sequential” proof, based on Theorem 341 and on Proposition 536, and give a proof based on the definition of limit.

Alternative proof Let $\varepsilon > 0$. We have to show that there exists $\delta > 0$ such that $f(x) \in V_\varepsilon(L) = (L - \varepsilon, L + \varepsilon)$ for every $x \in (U_\delta(x_0) - \{x_0\}) \cap A$. Since $\lim_{x \rightarrow x_0} g(x) = L$, there exists $\delta_1 > 0$ such that

$$x \in (U_{\delta_1}(x_0) - \{x_0\}) \cap A \implies L - \varepsilon < g(x) < L + \varepsilon \quad (12.33)$$

Since $\lim_{x \rightarrow x_0} h(x) = L$, there exists $\delta_2 > 0$ such that

$$x \in (U_{\delta_2}(x_0) - \{x_0\}) \cap A \implies L - \varepsilon < h(x) < L + \varepsilon \quad (12.34)$$

By taking $\delta = \min\{\delta_1, \delta_2\}$, both (12.33) and (12.34) then hold in $(U_\delta(x_0) - \{x_0\}) \cap A$. By (12.31), we then have

$$L - \varepsilon < g(x) \leq f(x) \leq h(x) < L + \varepsilon$$

for all $x \in (U_\delta(x_0) - \{x_0\}) \cap A$. That is,

$$f(x) \in V_\varepsilon(L)$$

for all $x \in (U_\delta(x_0) - \{x_0\}) \cap A$. Since ε was arbitrary, we conclude that $\lim_{x \rightarrow x_0} f(x) = L$. ■

*544 – Limit of the sum of functions

Proposition 544 Let $f, g : A \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ and $x_0 \in A'$, with $\lim_{x \rightarrow x_0} f(x) = L \in \overline{\mathbb{R}}$ and $\lim_{x \rightarrow x_0} g(x) = H \in \overline{\mathbb{R}}$.

(i) $\lim_{x \rightarrow x_0} (f + g)(x) = L + H$, provided that $L + H$ is not an indeterminate form (1.24), of the type

$$+\infty - \infty \quad \text{or} \quad -\infty + \infty$$

(ii) $\lim_{x \rightarrow x_0} (fg)(x) = LM$, provided that LM is not an indeterminate form (1.25), of the type

$$\pm\infty \cdot 0 \quad \text{or} \quad 0 \cdot (\pm\infty)$$

(iii) $\lim_{x \rightarrow x_0} (f/g)(x) = L/H$ provided that $g(x) \neq 0$ in a neighborhood of x_0 , with $x \neq x_0$, and that L/H is not an indeterminate form (1.26), of the type¹²

$$\frac{\pm\infty}{\pm\infty} \quad \text{or} \quad \frac{a}{0}$$

Proof We prove only (i), leaving to the reader the similar arguments for (ii) and (iii). Let $\{x_n\}$ be a sequence in A , with $x_n \neq x_0$ for every $n \geq 1$, such that $x_n \rightarrow x_0$. By Proposition 536, $f(x_n) \rightarrow L$ and $g(x_n) \rightarrow H$. Suppose that $L + H$ is not an indeterminate form. By Proposition 336, $(f + g)(x_n) \rightarrow L + H$. Therefore, by Proposition 536 it follows that $\lim_{x \rightarrow x_0} (f + g)(x) = L + H$. ■

*1059 – Tonelli's theorem

Theorem 1059 (Tonelli) A function $f : A \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ which is coercive and continuous on a subset C of A admits a maximizer in C , that is, there exists $\hat{x} \in C$ such that

$$f(\hat{x}) = \max_{x \in C} f(x)$$

If, in addition, C is closed, then $\arg \max_{x \in C} f(x)$ is compact.

Proof Since f is coercive, there exists $t \in \mathbb{R}$ such that the set $\Gamma = (f \geq t) \cap C$ is compact. By Weierstrass' Theorem, there exists $\hat{x} \in \Gamma$ such that $f(\hat{x}) \geq f(x)$ for every $x \in \Gamma$. At the same time, when $x \in C - \Gamma$ we have $f(x) < t$ and so $f(\hat{x}) \geq t > f(x)$. It follows that $f(\hat{x}) \geq f(x)$ for all $x \in C$, that is, $f(\hat{x}) = \max_{x \in C} f(x)$.

It remains to show that $\arg \max_{x \in C} f(x)$ is compact when C is closed. Since $\arg \max_{x \in C} f(x) \subseteq \Gamma$, it is enough to show that $\arg \max_{x \in C} f(x)$ is closed (in that a closed subset of a compact set is, in turn, compact). Clearly,

$$\arg \max_{x \in C} f(x) = \left(f \geq \max_{x \in C} f(x) \right) \cap C$$

In view of by Proposition 1053, we conclude that $\arg \max_{x \in C} f(x)$ is closed because it is the intersection of two closed sets. ■

Differential calculus

H1276 – Derivative of a constant

Example 1276 Consider a constant function $f : \mathbb{R} \rightarrow \mathbb{R}$, that is, $f(x) = k$ for every $x \in \mathbb{R}$. For every $h \neq 0$ we have

$$\frac{f(x+h) - f(x)}{h} = \frac{k - k}{h} = 0$$

and therefore $f'(x) = 0$ for every $x \in \mathbb{R}$. The derivative of a constant function is zero. ▲

*1286 – Derivability and continuity

Proposition 1286 A function $f : (a, b) \rightarrow \mathbb{R}$ derivable at a point $x \in (a, b)$ is continuous at x .

Proof We have to prove that $\lim_{x \rightarrow x_0} f(x) = f(x_0)$. Since f is derivable at x , the limit of the difference quotient exists and is finite, and it is equal to $f'(x_0)$:

$$\lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} = f'(x_0)$$

Let us rewrite the limit by setting $x = x_0 + h$, so that $h = x - x_0$. Observing that, as h tends to 0, we have that x tends to x_0 , we get:

$$\lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} = f'(x_0)$$

Therefore, by the algebra of limits (Proposition 336) we have:

$$\begin{aligned} \lim_{x \rightarrow x_0} (f(x) - f(x_0)) &= \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} (x - x_0) = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} \lim_{x \rightarrow x_0} (x - x_0) \\ &= f'(x_0) \lim_{x \rightarrow x_0} (x - x_0) = f'(x_0) \cdot 0 = 0 \end{aligned}$$

where the last equality holds since $f'(x_0)$ exists and is finite. We have thus proved that $\lim_{x \rightarrow x_0} (f(x) - f(x_0)) = 0$. On the other hand, again by algebra of limits, we have:

$$0 = \lim_{x \rightarrow x_0} (f(x) - f(x_0)) = \lim_{x \rightarrow x_0} f(x) - \lim_{x \rightarrow x_0} f(x_0) = \lim_{x \rightarrow x_0} f(x) - f(x_0)$$

Therefore $\lim_{x \rightarrow x_0} f(x) = f(x_0)$, as desired. ■

H1289 – Derivative of the exponential

Proposition 1289 The exponential function $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = \alpha^x$, with $\alpha > 0$, is derivable at each $x \in \mathbb{R}$, with derivative function $f' : \mathbb{R} \rightarrow \mathbb{R}$ given by

$$f'(x) = \alpha^x \log \alpha$$

In particular, $de^x/dx = e^x$, that is, the derivative function of the exponential function is the exponential function itself. So, the exponential function equals its derivative function, a truly remarkable invariance property that gives the exponential function a special status in differential calculus.

Proof We have

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\alpha^{x+h} - \alpha^x}{h} = \lim_{h \rightarrow 0} \frac{\alpha^x (\alpha^h - 1)}{h} \\ &= \alpha^x \lim_{h \rightarrow 0} \frac{\alpha^h - 1}{h} = \alpha^x \log \alpha \end{aligned}$$

where the last equality follows from the basic limit (12.40). ■

*1291 – Derivative of a linear combination

Proposition 1291 *Let $f, g : (a, b) \rightarrow \mathbb{R}$ be two derivable functions at $x \in (a, b)$. The sum function $f + g : (a, b) \rightarrow \mathbb{R}$ is derivable at x , with*

$$(f + g)'(x) = f'(x) + g'(x)$$

The result actually holds, more generally: for any linear combination $\alpha f + \beta g : (a, b) \rightarrow \mathbb{R}$, with $\alpha, \beta \in \mathbb{R}$, we have

$$(\alpha f + \beta g)'(x) = \alpha f'(x) + \beta g'(x) \quad (26.12)$$

In particular, the derivative of $\alpha f(x)$ is $\alpha f'(x)$.

Proof We prove the result directly in the more general form (26.12). We have

$$\begin{aligned} (\alpha f + \beta g)'(x) &= \lim_{h \rightarrow 0} \frac{(\alpha f + \beta g)(x+h) - (\alpha f + \beta g)(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(\alpha f)(x+h) + (\beta g)(x+h) - (\alpha f)(x) - (\beta g)(x)}{h} \\ &= \lim_{h \rightarrow 0} \left(\alpha \frac{f(x+h) - f(x)}{h} + \beta \frac{g(x+h) - g(x)}{h} \right) \\ &= \alpha \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} + \beta \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} \\ &= \alpha f'(x) + \beta g'(x) \end{aligned}$$

as desired. ■

H1300 – Derivative of log

Example 1300 Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the exponential function $f(x) = e^x$, so that $f^{-1} : (0, \infty) \rightarrow \mathbb{R}$ is the logarithmic function $f^{-1}(y) = \log y$. Given that $de^x/dx = e^x = y$, we have

$$\frac{d \log y}{dy} = \frac{1}{f'(x)} = \frac{1}{e^x} = \frac{1}{e^{\log y}} = \frac{1}{y}$$

for every $y > 0$. ▲

H1301 – Derivative of power functions

Proposition 1301 *The power function $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = x^a$, with $a \in \mathbb{R}$, is derivable at each $x \in \mathbb{R}$, with derivative function $f' : \mathbb{R} \rightarrow \mathbb{R}$ given by*

$$f'(x) = ax^{a-1}$$

Proof We have

$$x^a = e^{\log x^a} = e^{a \log x} \quad (26.21)$$

Setting $f(x) = e^x$ and $g(x) = a \log x$, from (26.21) it follows that

$$\frac{d(x^a)}{dx} = f'(g(x)) g'(x) = e^{a \log x} \frac{a}{x} = x^a \frac{a}{x} = ax^{a-1}$$

as desired. ■

H1303 – Derivative of arctan

Example 1303 Let $f : [-\pi/2, \pi/2] \rightarrow \mathbb{R}$ be given by $f(x) = \tan x$, so that $f^{-1} : \mathbb{R} \rightarrow [-\pi/2, \pi/2]$ is given by $f^{-1}(y) = \arctan y$. From (26.20) we have

$$\frac{d \tan x}{dx} = 1 + \tan^2 x = 1 + y^2$$

and so, for every $y \in \mathbb{R}$,

$$\frac{d \arctan y}{dy} = \frac{1}{1 + y^2}$$
▲

H1308 – Differentiability and derivability

Theorem 1308 *A function $f : (a, b) \rightarrow \mathbb{R}$ is differentiable at $x_0 \in (a, b)$ if and only if it is derivable at this point. In this case, the differential $df(x_0) : \mathbb{R} \rightarrow \mathbb{R}$ is given by*

$$df(x_0)(h) = f'(x_0)h$$

The differential at a point can be thus written in terms of the derivative at that point. *Inter alia*, this also shows the uniqueness of the differential $df(x_0)$.

Proof “If”. Let f be a function derivable at $x_0 \in (a, b)$. We have

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0) - f'(x_0)h}{h} &= \lim_{h \rightarrow 0} \left(\frac{f(x_0 + h) - f(x_0)}{h} - f'(x_0) \right) \\ &= \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} - f'(x_0) = 0 \end{aligned}$$

that is $f(x_0 + h) - f(x_0) - f'(x_0)h = o(h)$. Setting $m = f'(x_0)$, this implies (26.24) and therefore f is differentiable at x_0 .

“Only if”. Let f be differentiable at $x_0 \in (a, b)$. By (26.24),

$$f(x_0 + h) - f(x_0) = l(h) + o(h) \quad \text{as } h \rightarrow 0$$

The linear function $l : \mathbb{R} \rightarrow \mathbb{R}$ is a straight line passing through the origin, so there exists $m \in \mathbb{R}$ such that $l(h) = mh$. Hence

$$\lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} = \lim_{h \rightarrow 0} \frac{l(h) + o(h)}{h} = m \in \mathbb{R}$$

at x_0 the limit of the difference quotient exists and is finite and therefore f is derivable at x_0 . ■

H1365 – Fermat's theorem

Theorem 1365 (Fermat) *Let $f : A \subseteq \mathbb{R} \rightarrow \mathbb{R}$ be defined on a set A in $\mathbb{R}$ and C a subset of A . Let f be differentiable at an interior point $\hat{x}$ of C . If $\hat{x}$ is a local extremal point (a maximizer or a minimizer) of f on C , then*

$$f'(\hat{x}) = 0 \quad (28.1)$$

Proof Let $\hat{x} \in C$ be an interior point and a local maximizer on C (a similar argument holds if it is a local minimizer). There exists therefore $B_\varepsilon(\hat{x})$ such that (22.36) holds, that is, $f(\hat{x}) \geq f(x)$ for every $x \in B_\varepsilon(\hat{x}) \cap C$. For every $h > 0$ sufficiently small, that is, $h \in (0, \varepsilon)$, we have $\hat{x} + h \in B_\varepsilon(\hat{x})$. Hence

$$\frac{f(\hat{x} + h) - f(\hat{x})}{h} \leq 0 \quad \forall h \in (0, \varepsilon)$$

which implies

$$\lim_{h \rightarrow 0} \frac{f(\hat{x} + h) - f(\hat{x})}{h} = \lim_{h \rightarrow 0^+} \frac{f(\hat{x} + h) - f(\hat{x})}{h} \leq 0 \quad (28.2)$$

where the limits exist and are equal because f is differentiable at $\hat{x}$.

On the other hand, for every $h < 0$ sufficiently small, that is, $h \in (-\varepsilon, 0)$, we have $\hat{x} + h \in B_\varepsilon(\hat{x})$. Therefore,

$$\frac{f(\hat{x} + h) - f(\hat{x})}{h} \geq 0 \quad \forall h \in (-\varepsilon, 0)$$

which implies

$$\lim_{h \rightarrow 0} \frac{f(\hat{x} + h) - f(\hat{x})}{h} = \lim_{h \rightarrow 0^-} \frac{f(\hat{x} + h) - f(\hat{x})}{h} \geq 0 \quad (28.3)$$

where, again, the limits exist and are equal because f is differentiable at $\hat{x}$.

Together, inequalities (28.2) and (28.3) imply

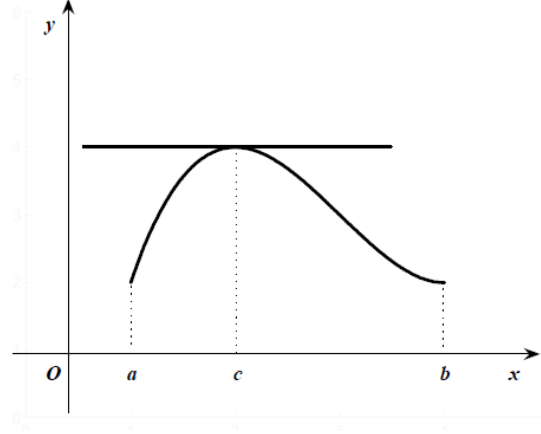
$$f'(\hat{x}) = \lim_{h \rightarrow 0} \frac{f(\hat{x} + h) - f(\hat{x})}{h} = 0$$

as desired. ■

H1374 – Rolle's theorem

Theorem 1374 (Rolle) Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$, with $f(a) = f(b)$, and differentiable on (a, b) . Then, there exists (at least) one critical point $\hat{x} \in (a, b)$, that is, a point $\hat{x} \in (a, b)$ such that $f'(\hat{x}) = 0$.

This theorem, which provides a simple sufficient condition for a function to have a critical point, has an immediate graphical intuition:



Proof By Weierstrass' Theorem, there exist $x_1, x_2 \in [a, b]$ such that $f(x_1) = \min_{x \in [a, b]} f(x)$ and $f(x_2) = \max_{x \in [a, b]} f(x)$. Denote $m = \min_{x \in [a, b]} f(x)$ and $M = \max_{x \in [a, b]} f(x)$. If $m = M$, then f is constant, that is, $f(x) = m = M$, and therefore $f'(x) = 0$ for every $x \in (a, b)$. If $m < M$, then at least one of the points x_1 and x_2 is interior to $[a, b]$. Indeed, they cannot be both boundary points because $f(a) = f(b)$. If x_1 is an interior point of $[a, b]$, that is, $x_1 \in (a, b)$, then by Fermat's Theorem we have $f'(x_1) = 0$, so $\hat{x} = x_1$. Analogously, if $x_2 \in (a, b)$, we have $f'(x_2) = 0$, and therefore $\hat{x} = x_2$. ■

H1376 – Mean value theorem

Theorem 1376 (Mean Value) Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b) . Then, there exists $\hat{x} \in (a, b)$ such that

$$f'(\hat{x}) = \frac{f(b) - f(a)}{b - a} \quad (28.7)$$

Proof Let $g : [a, b] \rightarrow \mathbb{R}$ be the auxiliary function defined by

$$g(x) = f(x) - \left(f(a) + \frac{f(b) - f(a)}{b - a} (x - a) \right)$$

It is the difference between f and the straight line passing through the points $(a, f(a))$ and $(b, f(b))$. The function g is continuous on $[a, b]$ and differentiable on (a, b) . Moreover, $g(a) = g(b) = 0$. By Rolle's Theorem, there exists $\hat{x} \in (a, b)$ such that $g'(\hat{x}) = 0$. But

$$g'(x) = f'(x) - \frac{f(b) - f(a)}{b - a}$$

and therefore

$$f'(\hat{x}) - \frac{f(b) - f(a)}{b - a} = 0$$

That is, $\hat{x}$ satisfies condition (28.7). ■

H1377 – Characterization of functions with null derivative

Corollary 1377 *Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b) . Then $f'(x) = 0$ for every $x \in (a, b)$ if and only if f is constant, that is, if and only if there exists $k \in \mathbb{R}$ such that*

$$f(x) = k \quad \forall x \in [a, b]$$

Proof Let us prove the “only if”, since the “if” is the simple property of derivatives seen in Example 1276. Let $x \in (a, b)$ and let us apply the Mean Value Theorem on the interval $[a, x]$. It yields a point $\hat{x} \in (a, x)$ such that

$$0 = f'(\hat{x}) = \frac{f(x) - f(a)}{x - a}$$

that is, $f(x) = f(a)$. Since x is any point in $(a, b]$, it follows that $f(x) = f(a)$ for any $x \in [a, b]$. ■

H1378 – Characterization of functions with the same derivative

Corollary 1378 *Let $f, g : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b) . Then $f'(x) = g'(x)$ for every $x \in (a, b)$ if and only if there exists $k \in \mathbb{R}$ such that*

$$f(x) = g(x) + k \quad \forall x \in [a, b]$$

Two functions that have the same first derivative are, thus, equal up to an (additive) constant k .

Proof Here too we prove the “only if”, the “if” being obvious. Let $h : [a, b] \rightarrow \mathbb{R}$ be the auxiliary function $h(x) = f(x) - g(x)$. We have $h'(x) = f'(x) - g'(x) = 0$ for every $x \in (a, b)$. Therefore, by Corollary 1377 h is constant on $[a, b]$. That is, there exists $k \in \mathbb{R}$ such that $h(x) = k$ for every $x \in [a, b]$, so $f(x) = g(x) + k$ for every $x \in [a, b]$. ■

*1387 – Monotonicity test on an interval

Proposition 1387 Let $f : (a, b) \rightarrow \mathbb{R}$ be a differentiable function, with $a, b \in \overline{\mathbb{R}}$. Then, f is (globally) increasing on (a, b) if and only if $f'(x) \geq 0$ for every $x \in (a, b)$.

Because of the clause $a, b \in \overline{\mathbb{R}}$, the interval (a, b) can be unbounded, for example $(a, b) = \mathbb{R}$. A dual result, with negativity of the derivative on (a, b) , holds for the decreasing monotonicity. Note that Corollary 1377 is then a special case since $f'(x) = 0$ for every $x \in (a, b)$ is equivalent to having both $f'(x) \geq 0$ and $f'(x) \leq 0$ for every $x \in (a, b)$ and therefore, being simultaneously increasing and decreasing, f is constant.

Proof “Only if”. Suppose that f is increasing. Let $x \in (a, b)$. For every $h > 0$ we have $f(x+h) \geq f(x)$, hence

$$\frac{f(x+h) - f(x)}{h} \geq 0$$

It follows that

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0^+} \frac{f(x+h) - f(x)}{h} \geq 0$$

“If”. Let $f'(x) \geq 0$ for every $x \in (a, b)$. Let $x_1, x_2 \in (a, b)$ with $x_1 < x_2$. By the Mean Value Theorem, there exists $\hat{x} \in [x_1, x_2]$ such that

$$f'(\hat{x}) = \frac{f(x_2) - f(x_1)}{x_2 - x_1} \quad (28.13)$$

Since $f'(\hat{x}) \geq 0$ and $x_2 - x_1 > 0$, this shows that $f(x_2) \geq f(x_1)$. ■

H1389 – Strict monotonicity test on an interval

Proposition 1389 Let $f : (a, b) \rightarrow \mathbb{R}$ be a differentiable function, with $a, b \in \overline{\mathbb{R}}$. If $f'(x) > 0$ for every $x \in (a, b)$, then f is (globally) strictly increasing on (a, b) .

Proof The proof is similar to that of Proposition 1387 and is a simple application of the Mean Value Theorem. Let $f'(x) > 0$ for every $x \in (a, b)$ and let $x_1, x_2 \in (a, b)$ with $x_1 < x_2$. By the Mean Value Theorem, there exists $c \in [x_1, x_2]$ such that

$$f'(c) = \frac{f(x_2) - f(x_1)}{x_2 - x_1} \quad (28.14)$$

Since $f'(c) > 0$ for every c and $x_2 - x_1 > 0$, from (28.14) it follows that $f(x_2) > f(x_1)$. ■

H1427 – Maclaurin formula for $\log(1+x)$

Example 1427 The function $f : (-1, \infty) \rightarrow \mathbb{R}$ given by $f(x) = \log(1+x)$ is n times differentiable at each point of its domain, with

$$f^{(n)}(x) = (-1)^{n+1} \frac{(n-1)!}{(1+x)^n} \quad \forall n \geq 1 \quad (29.17)$$

Therefore, Taylor's expansion of order n of f at $x_0 > -1$ is

$$\begin{aligned} \log(1+x_0+h) &= \log(1+x_0) + \frac{h}{1+x_0} - \frac{h^2}{2(1+x_0)^2} \\ &\quad + \frac{h^3}{3(1+x_0)^3} + \cdots + (-1)^{n+1} \frac{h^n}{n(1+x_0)^n} + o(h^n) \\ &= \log(1+x_0) + \sum_{k=1}^n (-1)^{k+1} \frac{h^k}{k(1+x_0)^k} + o(h^n) \end{aligned}$$

or equivalently, using (29.16),

$$\log(1+x) = \log(1+x_0) + \sum_{k=1}^n (-1)^{k+1} \frac{(x-x_0)^k}{k(1+x_0)^k} + o((x-x_0)^n)$$

A simple polynomial thus locally approximates the logarithmic function. In particular, the Maclaurin's expansion of order n of f is

$$\begin{aligned} \log(1+x) &= x - \frac{x^2}{2} + \frac{x^3}{3} + \cdots + (-1)^{n+1} \frac{x^n}{n} + o(x^n) \\ &= \sum_{k=1}^n (-1)^{k+1} \frac{x^k}{k} + o(x^n) \end{aligned} \quad (29.18)$$

▲

H1428 - Maclaurin formulae for special functions

Example 1428 In a similar way the reader can verify the Maclaurin's expansions of order n of the following elementary functions:

$$\begin{aligned} e^x &= 1 + x + \frac{x^2}{2} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!} + o(x^n) = \sum_{k=0}^n \frac{x^k}{k!} + o(x^n) \\ \sin x &= x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 + \cdots + \frac{(-1)^n}{(2n+1)!}x^{2n+1} + o(x^{2n+1}) = \sum_{k=0}^n \frac{(-1)^k}{(2k+1)!}x^{2k+1} + o(x^{2n+1}) \\ \cos x &= 1 - \frac{1}{2}x^2 + \frac{1}{4!}x^4 + \cdots + \frac{(-1)^n}{(2n)!}x^{2n} + o(x^{2n}) = \sum_{k=0}^n \frac{(-1)^k}{(2k)!}x^{2k} + o(x^{2n}) \end{aligned}$$

Here too it is important to observe how these functions can be locally (well) approximated by simple polynomials. ▲

H1368 – Fermat's theorem in $\mathbb{R}^n$

Theorem 1368 *Let $f : A \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ be defined on a set A in $\mathbb{R}^n$ and C a subset of A . Suppose f is differentiable at an interior point $\hat{x}$ of C . If $\hat{x}$ is a local extremal point (a maximizer or a minimizer) of f on C , then*

$$\nabla f(\hat{x}) = \mathbf{0} \quad (28.4)$$

We leave the proof to the reader. Indeed, *mutatis mutandis*, it is the same as that of Fermat's Theorem.³

Linear algebra

*69 – The intersection of vector subspaces

Proposition 69 *The intersection of any collection of vector subspaces of $\mathbb{R}^n$ is a vector subspace.*

Proof Let $\{V_i\}$ be any collection of vector subspaces of $\mathbb{R}^n$. Since $\mathbf{0} \in V_i$ for every i , we have $\bigcap_i V_i \neq \emptyset$. Let $x, y \in \bigcap_i V_i$ and $\alpha, \beta \in \mathbb{R}$. Since $x, y \in \bigcap_i V_i$, we have $x, y \in V_i$ for every i and, therefore, $\alpha x + \beta y \in V_i$ for every i since each V_i is a vector subspace of $\mathbb{R}^n$. Hence, $\alpha x + \beta y \in \bigcap_i V_i$, and so $\bigcap_i V_i$ is a vector subspace of $\mathbb{R}^n$. ■

*84 – Span and linear combinations

Theorem 84 *Let S be a set of $\mathbb{R}^n$. A vector $x \in \mathbb{R}^n$ belongs to $\text{span } S$ if and only if it is a linear combination of vectors of S .*

Proof We need to prove that $x \in \mathbb{R}^n$ belongs to $\text{span } S$ if and only if there exist a finite set $\{x^i\}_{i \in I}$ of vectors in S and a finite set $\{\alpha_i\}_{i \in I}$ of scalars such that $x = \sum_{i \in I} \alpha_i x^i$. “If”. Let $x \in \mathbb{R}^n$ be a linear combination of a finite set $\{x^i\}_{i \in I}$ of vectors of S . For simplicity, set $\{x^i\}_{i \in I} = \{x^1, \dots, x^k\}$. There exists, therefore, a set $\{\alpha_i\}_{i=1}^k$ of real numbers such that $x = \sum_{i=1}^k \alpha_i x^i$. By the definition of vector subspace, we have $\alpha_1 x^1 + \alpha_2 x^2 \in \text{span } S$ since $x^1, x^2 \in \text{span } S$. In turn, $(\alpha_1 x^1 + \alpha_2 x^2) \in \text{span } S$ implies $(\alpha_1 x^1 + \alpha_2 x^2) + \alpha_3 x^3 \in \text{span } S$, and by proceeding in this way we get that $x = \sum_{i=1}^k \alpha_i x^i \in \text{span } S$, as claimed.

“Only if”. Let V be the set of all vectors $x \in \mathbb{R}^n$ that can be expressed as linear combinations of vectors of S , that is, $x \in V$ if there exist finite sets $\{x^i\}_{i \in I} \subseteq S$ and $\{\alpha^i\}_{i \in I} \subseteq \mathbb{R}$ such that $x = \sum_{i \in I} \alpha_i x^i$. It is easy to see that V is a vector subspace of $\mathbb{R}^n$ containing S . It follows that $\text{span } S \subseteq V$ and so each $x \in \text{span } S$ is a linear combination of vectors of S . ■

*90 – Unique writing of a basis

Theorem 90 *A finite subset S of $\mathbb{R}^n$ is a basis of $\mathbb{R}^n$ if and only if each vector $x \in \mathbb{R}^n$ can be written in only one way as a linear combination of vectors in S .*

Proof “Only if”. Let $S = \{x^i\}_{i=1}^m$ be a basis of $\mathbb{R}^n$. By Theorem 84 and since $\text{span } S = \mathbb{R}^n$, each vector $x \in \mathbb{R}^n$ can be represented as a linear combination of elements of S . Given $x \in \mathbb{R}^n$, suppose that there exist two sets of scalars $\{\alpha_i\}_{i=1}^m$ and $\{\beta_i\}_{i=1}^m$ such that

$$x = \sum_{i=1}^m \alpha_i x^i = \sum_{i=1}^m \beta_i x^i$$

Hence,

$$\sum_{i=1}^m (\alpha_i - \beta_i) x^i = \mathbf{0}$$

and, since the vectors in S are linearly independent, it follows that $\alpha_i - \beta_i = 0$ for every $i = 1, \dots, m$. That is, $\alpha_i = \beta_i$ for every $i = 1, \dots, m$.

“If”. Let $S = \{x^1, \dots, x^m\}$ and suppose that each $x \in \mathbb{R}^n$ can be written in a unique way as a linear combination of vectors in S . Clearly, by Theorem 84 we have $\mathbb{R}^n = \text{span } S$. It

remains to prove that S is a linearly independent set. Suppose that the scalars $\{\alpha_i\}_{i=1}^m$ are such that

$$\sum_{i=1}^m \alpha_i x^i = \mathbf{0}$$

Since we also have

$$\sum_{i=1}^m 0x^i = \mathbf{0}$$

we conclude that $\alpha_i = 0$ for every $i = 1, \dots, m$ because, by hypothesis, the vector $\mathbf{0}$ can be written in only one way as a linear combination of vectors in S . ■

H119 – Fourier coefficients

Proposition 119 *Let $\{x^1, \dots, x^n\}$ be an orthonormal basis of $\mathbb{R}^n$. For every $y \in \mathbb{R}^n$, we have*

$$y = \sum_{i=1}^n (y \cdot x^i) x^i \quad (4.13)$$

The coefficients $y \cdot x^i$ are called *Fourier coefficients* of y with respect to the orthonormal basis $\{x^1, \dots, x^n\}$.

Proof Since $\{x^1, \dots, x^n\}$ is a basis, there exist n scalars $\alpha_1, \alpha_2, \dots, \alpha_n$ such that

$$y = \sum_{i=1}^n \alpha_i x^i$$

For $j = 1, 2, \dots, n$ the scalar product $y \cdot x^j$ is

$$y \cdot x^j = \sum_{i=1}^n \alpha_i (x^i \cdot x^j)$$

Since $\{x^1, \dots, x^n\}$ is orthonormal, we have

$$x^i \cdot x^j = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j \end{cases}$$

Hence $y \cdot x^j = \alpha_j$, from which the statement follows. ■

H669 – Riesz' representation theorem

Theorem 669 (Riesz) *A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is linear if and only if there exists a unique vector $\chi \in \mathbb{R}^n$ such that*

$$f(x) = \chi \cdot x \quad \forall x \in \mathbb{R}^n$$

Proof We have already seen the “if” part in Example 661. It remains to prove the “only if” part. So, let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a linear function and consider the standard basis $\{e^1, \dots, e^n\}$ of $\mathbb{R}^n$. Set

$$\chi = (f(e^1), \dots, f(e^n)) \in \mathbb{R}^n$$

As seen in (4.15), we can write each vector $x \in \mathbb{R}^n$ as $x = \sum_{i=1}^n x_i e^i$. By equality (15.3), we thus have:

$$f(x) = f\left(\sum_{i=1}^n x_i e^i\right) = \sum_{i=1}^n x_i f(e^i) = \sum_{i=1}^n \chi_i x_i = \chi \cdot x \quad \forall x \in \mathbb{R}^n$$

As to the uniqueness, let $\chi, \chi' \in \mathbb{R}^n$ be vectors such that both $f(x) = \chi \cdot x$ and $f(x) = \chi' \cdot x$ for all $x \in \mathbb{R}^n$. For every $i = 1, \dots, n$ we then have

$$\chi'_i = \chi' \cdot e^i = f(e^i) = \chi \cdot e^i = \chi_i$$

and so $\chi' = \chi$. ■

*698 – Representation theorem

Theorem 698 (Operator Riesz Theorem) *An operator $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is linear if and only if there exists a unique matrix $A_{m \times n}$ such that*

$$T(x) = Ax \quad \forall x \in \mathbb{R}^n \tag{15.12}$$

The matrix A is called *matrix associated to the operator T* (or also *representative matrix of the operator T*). It reduces to the vector $\chi \in \mathbb{R}^n$ of Riesz's Theorem, which is the special case of this theorem when $m = 1$.

Proof “If”. This part is contained, essentially, in Example 689. “Only if”. Let T be a linear operator. Set

$$A = \begin{bmatrix} T(e^1) & T(e^2) & \dots & T(e^n) \\ m \times 1 & m \times 1 & & m \times 1 \end{bmatrix} \quad (15.13)$$

that is, A is the $m \times n$ matrix whose n columns are the column vectors $T(e^j)$ for $j = 1, \dots, n$.⁹ As seen in (4.15), we can write every vector $x \in \mathbb{R}^n$ as $x = \sum_{j=1}^n x_j e^j$. Therefore, for every $x \in \mathbb{R}^n$,

$$\begin{aligned} T(x) &= T\left(\sum_{j=1}^n x_j e^j\right) = \sum_{j=1}^n x_j T(e^j) \\ &= x_1 \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix} + x_2 \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix} + \dots + x_n \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix} \\ &= \begin{bmatrix} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n \end{bmatrix} = \begin{bmatrix} a^1 \cdot x \\ a^2 \cdot x \\ \vdots \\ a^m \cdot x \end{bmatrix} = Ax \end{aligned}$$

where $a^1, a^2, \dots, a^m$ are the rows of A .

As to uniqueness, let A and B be two $m \times n$ matrices such that both $T(x) = Ax$ and $T(x) = Bx$ for all $x \in \mathbb{R}^n$. By considering the vectors e^j we have

$$\begin{aligned} (a_{11}, a_{21}, \dots, a_{m1}) &= Ae^1 = T(e^1) = Be^1 = (b_{11}, b_{21}, \dots, b_{m1}) \\ (a_{12}, a_{22}, \dots, a_{m2}) &= Ae^2 = T(e^2) = Be^2 = (b_{12}, b_{22}, \dots, b_{m2}) \\ &\vdots \\ (a_{1n}, a_{2n}, \dots, a_{mn}) &= Ae^n = T(e^n) = Be^n = (b_{1n}, b_{2n}, \dots, b_{mn}) \end{aligned}$$

Therefore, $A = B$. ■

*+H705 – Kernel and image are subspaces

Lemma 705 *If $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is linear, then $\ker T$ is a vector subspace of $\mathbb{R}^n$ and $\text{Im } T$ is a vector subspace of $\mathbb{R}^m$.*

Proof We show the result for $\ker T$, leaving $\text{Im } T$ to the reader. Let $x, y \in \ker T$, i.e., $T(x) = \mathbf{0}$ and $T(y) = \mathbf{0}$. For every $\alpha, \beta \in \mathbb{R}$, we have

$$T(\alpha x + \beta y) = \alpha T(x) + \beta T(y) = \alpha \mathbf{0} + \beta \mathbf{0} = \mathbf{0}$$

Thus, $\alpha x + \beta y \in \ker T$ and this proves that $\ker T$ is a vector subspace of $\mathbb{R}^n$. ■

H – Determinant of the inverse matrix

Theorem 748 (Binet) *If A and B are two square matrices of the same order n , then*

$$\det AB = \det A \det B$$

So, determinants commute, i.e., $\det AB = \det BA$. This is a first interesting consequence of Binet's Theorem. Since $I = A^{-1}A$, another interesting consequence of this result is the formula

$$\det A^{-1} = \frac{1}{\det A}$$

when A is invertible. Indeed, $1 = \det I = \det A^{-1}A = \det A^{-1} \det A$. In particular, this formula implies that

$$\det A \neq 0 \iff \det A^{-1} \neq 0$$

H765 – Cramer's theorem

Theorem 765 (Cramer) *Let A be a square matrix of order n . The linear system*

$$Ax = b$$

has a unique solution for every $b \in \mathbb{R}^n$ if and only if the matrix A is invertible. In this case, the solution is given by

$$x = A^{-1}b \tag{15.41}$$

*772 – Kronecker-Capelli theorem

Theorem 772 (Kronecker-Capelli) *The linear system (15.48) is solvable if and only if $\rho(A) = \rho(A|b)$.*

Proof The linear system (15.48) is solvable if and only if $b \in \text{Im } T$. Since $\text{Im } T$ is the vector subspace of $\mathbb{R}^m$ generated by the columns of A , the system is solvable if and only if b is a linear combination of such columns. That is, if and only if the matrices A and $A|b$ have the same number of linearly independent columns, so the same rank. ■